BAMBARE

TRAVEL MANAGEMENT SYSTEM

INTRODUCTION

The tourism industry requires efficient digital platforms for travel planning, booking, and customer management. Currently, tourists use multiple applications to search for travel packages, manage bookings, and track expenses, leading to fragmented experiences.

Bambare is proposed as a centralized Travel Management System that integrates travel packages, bookings, navigation, and customer information into a single platform, improving convenience for tourists and management efficiency for administrators.

Project Overview

Bambare is a cross-platform Travel Management System designed to support both tourists and travel agency administrators.

- The system provides a unified platform for searching, comparing, booking, and managing travel arrangements including flights, hotels, car rentals, and activities.
- It includes real-time availability and pricing updates from travel service providers.
- Users can manage their travel itineraries, receive booking confirmations, and monitor their travel plans.
- Administrators can manage customers, bookings, and system data through a dashboard and generate reports.

Problem Statements and Problem Solutions

Problem Statements

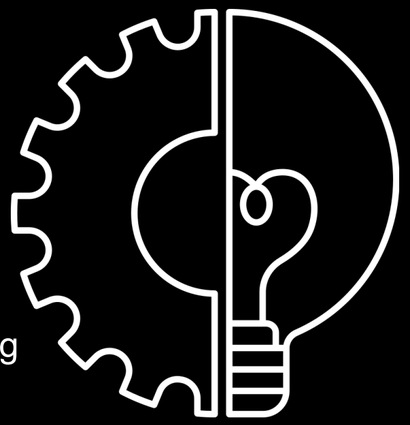
Tourists use multiple platforms for different travel activities.

Booking and travel management processes are fragmented and time-consuming.

Lack of centralized data causes inefficiency in travel management.

Travelers face difficulties in tracking bookings and receiving real-time updates.

Existing systems offer limited administrative control and reporting.



Problem Solutions

Develop a centralized Travel Management System that integrates all travel services.

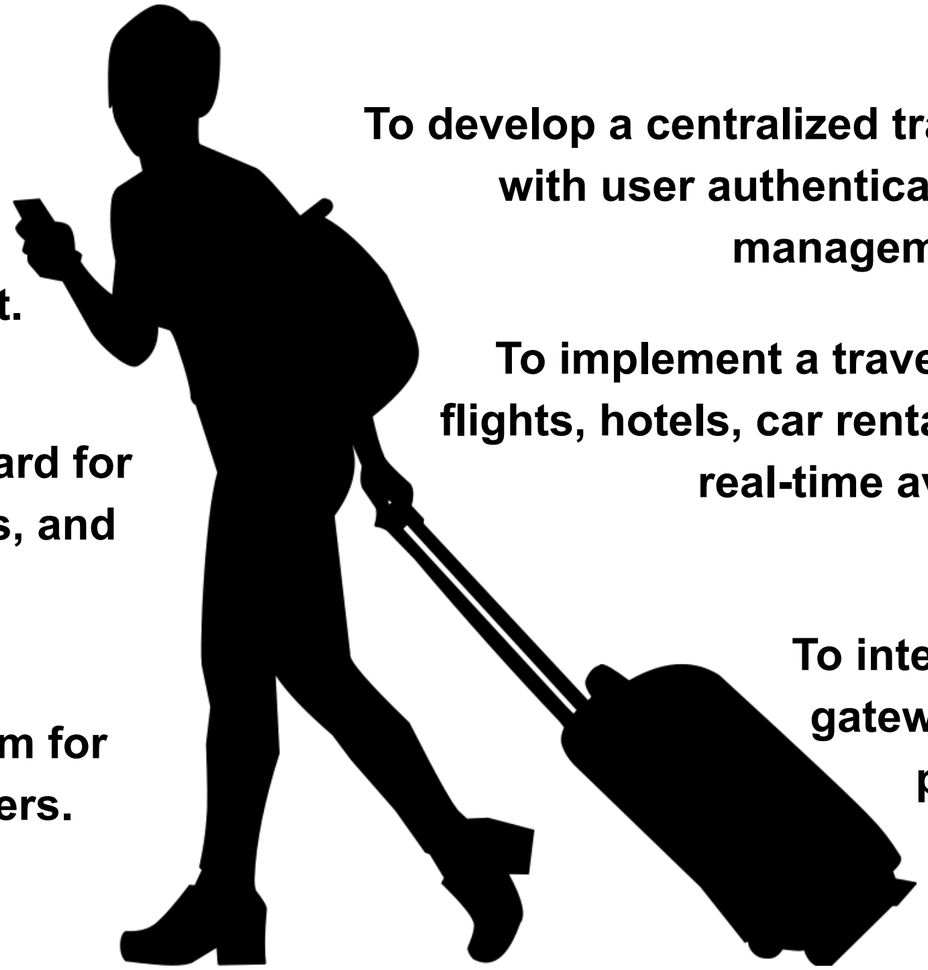
Provide real-time availability and pricing information.

Enable automated booking confirmations and itinerary management.

Implement a complete admin dashboard for system monitoring and reporting.

Improve customer experience by reducing booking time and providing accurate information.

Project Objectives



To develop a centralized travel booking platform with user authentication and profile management.

To provide automated booking confirmations and itinerary management.

To implement a travel search engine for flights, hotels, car rentals, and activities with real-time availability.

To implement an admin dashboard for managing bookings, customers, and generating reports.

To integrate a secure payment gateway supporting multiple payment methods.

To support a notification system for booking updates and reminders.

Key Features of Bambare

- User registration, login, and profile management
- Integrated search for hotels, car rentals, and activities
- Real-time availability and pricing updates
- Secure payment gateway integration
- Automated booking confirmation and itinerary generation
- Map-based navigation, destination and activity recommendations
- Expense tracking with currency conversion
- Admin dashboard with reporting and analytics
- Notification system (email/SMS)

System Users

Tourists

Search and compare travel packages

Book hotels, car rentals, and activities

Manage itineraries and receive booking confirmations

Track travel expenses and view summaries

Use navigation and view recommended places

Administrators

Manage customers, bookings, and system data

Monitor bookings, revenue, and customer insights

Send notifications and updates

Generate reports and analytics

Technology Stack

Frontend
Cross-platform mobile framework (Android & iOS)

Backend
Firebase

Database
Firestore

Authentication
Firebase Authentication

Maps & Navigation
Google Maps API

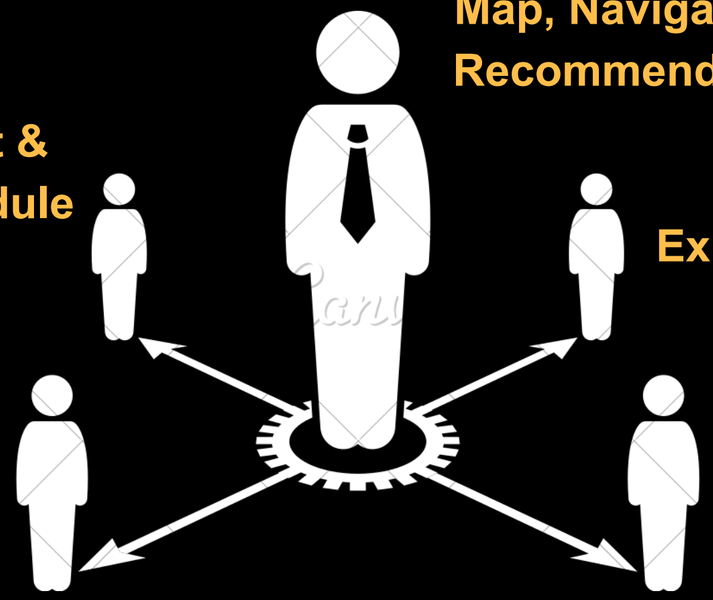
Notifications
Firebase Cloud Messaging

Payment Gateway
(To be integrated with a supported provider)

Travel APIs
Integration with major travel service providers

Function Distribution

The project is distributed among five specialized team members, each owning a complete functional module to ensure efficient development, clear accountability, and seamless integration.



Map, Navigation & Places Recommendation Module

User Management & Authentication Module

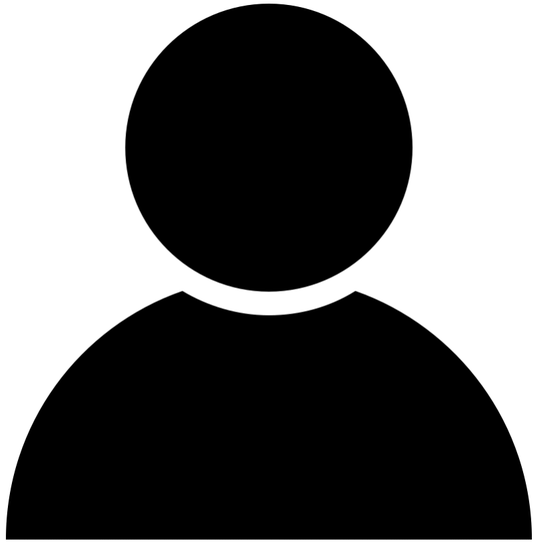
Expense Tracking & Currency Converter Module

Admin Panel, Notifications & Reporting Module

Tour & Activity Planning and Booking Module

Function Distribution Handled by Avishka

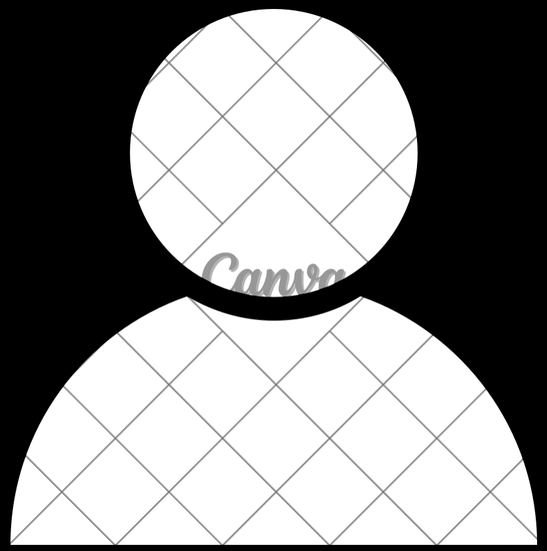
Map, Navigation & Places Recommendation Module



- Integrate Google Maps API for real-time navigation and location services
- Enable location pinning and map-based destination marking
- Provide navigation support between selected locations
- Manage recommended destinations and activities through CRUD operations
- Access and manage location data in the database
- Generate reports on popular destinations and user travel trends

Function Distribution Handled by Prarthana

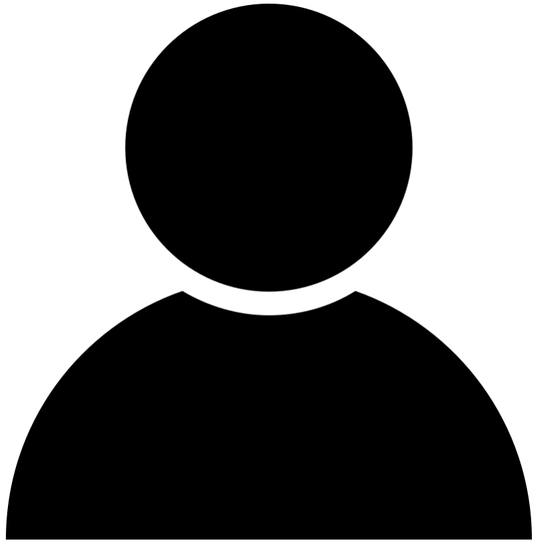
Expense Tracking & Currency Converter Module



- Design and implement the expense tracking interface
- Manage Firestore transactions for financial data
- Add, update, view, and delete travel expenses
- Provide currency conversion using API or static exchange rates
- Generate monthly expense reports for users

Function Distribution Handled by Shamla

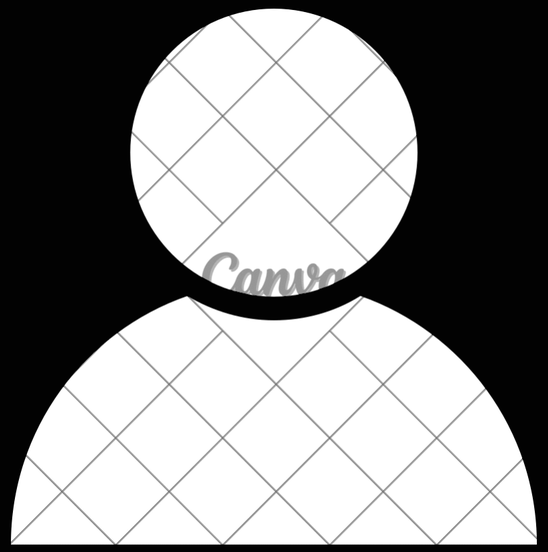
Tour & Activity Planning and Booking Module



- Design UI for tour and activity listings
- Enable search, selection, and booking of tours and activities
- Perform CRUD operations for tours and booking records
- Access booking data using Firestore database
- Validate availability before confirming bookings
- Generate booking summaries and reports

Function Distribution Handled by Vidhuja

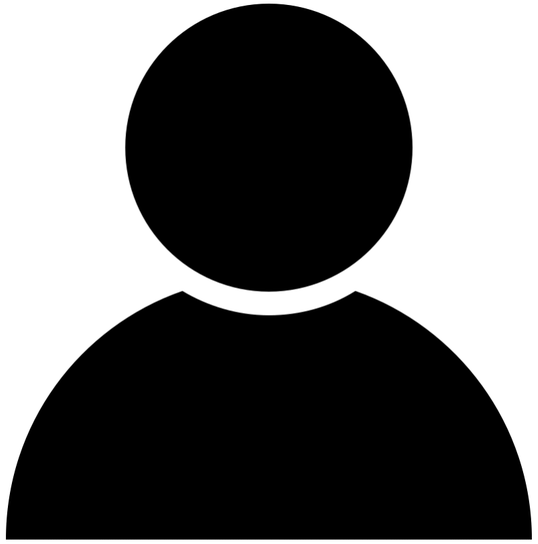
User Management & Authentication Module



- Manage user profiles using CRUD operations
- Maintain database connectivity for user data
- Implement Firebase authentication for secure access
- Manage user roles (Admin/User)
- Design login and registration UI
- Generate user registration and activity reports

Function Distribution Handled by Oshidi

Admin Panel, Notifications & Reporting Module



- Design and implement the admin dashboard interface
- Perform CRUD operations for hotels, tours, and users
- Implement Firebase Cloud Messaging for notifications
- Generate system-wide reports for bookings, users, and revenue
- Control database access and permissions

Conclusion

Bambare provides a centralized Travel Management System that simplifies travel planning, booking, navigation, and expense management. The system enhances the travel experience for tourists while enabling administrators to efficiently manage bookings, customer data, and generate reports. This solution meets the goals defined in the project charter and supports real-time availability and reliable system performance.

Total estimated budget	**22, 500.00 LKR**

Thank You

Travel is the only thing you buy that makes you richer